

# Jingqi Wang

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## Education:

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**Bachelor of Science, Computer Science, Mathematics**

**GPA: 3.83**

**May 2029**

Rose-Hulman Institute of Technology – Terre Haute, IN

*Junior Standing*

*Relevant Courses:* Computer Architecture, Operating Systems, Artificial Intelligence, Data Mining, Machine Learning, Database Systems, Design and Analysis of Algorithms, Programming Language Concepts, Web Programming

## Skills:

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*Programming Languages:* Python, Java, C, C++

*Assembly & HDL:* ARM assembly, RISC-V assembly, Verilog

*OS:* Linux

*Software:* MATLAB, Quartus Prime, Anaconda, Visual Studio Code

*Language:* English, Mandarin Chinese

*GitHub:* <https://github.com/agneswang42>

## Project Experience:

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### GraphSurfing (Java, Data Structures & Algorithms)

- Developed a generic graph framework in Java featuring adjacency-list and adjacency-matrix representations, efficient edge operations, and lazy neighbor iterators.
- Implemented graph algorithms including shortest-path search and strongly connected component detection using graph traversal techniques.
- Evaluated time and space complexity trade-offs between graph representations through performance testing and optimization.
- Applied solutions to a real-world Wikipedia hyperlink graph containing over 700,000 vertices and 3.6 million edges.

### Java Pixel Collision Game | Team Lead

- Led a team project to design and implement a Java GUI-based pixel collision game using object-oriented programming and UML design principles.
- Implemented core gameplay features, including player/enemy movement, collision detection, level progression, sound effects, and real-time score display.
- Presented project architecture and functionality during a formal class and faculty demonstration.

### FPGA Atari PONG Recreation

- Designed and implemented a modified Atari PONG arcade game on FPGA hardware using Verilog HDL and Altera Quartus Prime.
- Developed, simulated, synthesized, and deployed digital circuits while performing timing analysis and hardware validation.
- Produced technical documentation and presented project outcomes through reports, demonstrations, and presentations.